

Project Title
**Parking Demand Prediction Using Elastic Net
Regression**

Project Report submitted for the partial fulfilment of grade for the course

MACHINE LEARNING (25SC2107E)
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Submitted by

Team Number: 14

Section: 7

(2520030321) CHEEKATI CHARAN SAI

(2520030340) SRIKRUTHI

(2520030372) CHADA HASIKA

Under the guidance of
GADDAM RAVINDRA
BABU

Professor, Department of CSE



ABSTRACT

Urban areas experience a significant increase in vehicle usage, leading to high parking demand and traffic congestion. Finding available parking spaces has become a time-consuming and inefficient task, causing unnecessary fuel consumption, increased travel time, and environmental pollution. Accurate prediction of parking demand is essential for efficient parking management and smart city planning.

The **Parking Demand Prediction Using Elastic Net Regression** system is designed to predict future parking demand by analyzing historical parking data and various influencing factors such as time, day, location, weather conditions, and seasonal trends. The system enables users to collect, preprocess, and analyze parking data through an efficient prediction framework. To enhance prediction accuracy, Elastic Net Regression is employed as it combines the strengths of Ridge and Lasso regression techniques, effectively handling multicollinearity and performing feature selection. The proposed system processes historical data, trains the regression model, evaluates prediction performance, and generates reliable parking demand forecasts. By providing accurate predictions, the system helps optimize parking space allocation, reduce traffic congestion, minimize vehicle search time, improve parking utilization, and support intelligent decision-making for smart transportation and urban infrastructure management.

Signature of the Guide